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Ingestion Team

AI-Led Development

Environment

Claude Code + GitHub Copilot + New Relic MCP

🚀 Rapid Development

🔍 Instant Logs & RCA

⚡ Multi-Agent Orchestration

🧠 Compound Knowledge

📈 3× Productivity

The Challenge — Four Real Pains the DataOS Ingestion Team Was Living With

🕐 Slow Development Tickets

2–3 DAYS

per ticket — regardless of complexity or volume

🔍 Pipeline Debugging

30 MIN–3 HRS

backtracking alert issues one by one manually

💥 Hidden Blast Radius

5-DAY

incident caused by a change no one knew would break

🗂️ No Historical Context

HOURS–DAYS

re-discovering repo structure and past decisions
every time someone joined, left, or came back

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DataOS Ingestion — What We Own & Tools in Use

What We Own — Data Pipeline

DATA SOURCES

Billing Platform · Kafka Connect · SFDC · Zuora · Google Sheets · etc.

↓

INGESTION — WE OWN THIS

Fivetran · Snowflake · Iceberg · Glue
Spark · NRDB · Custom Pipelines

↓

ZONE 2 DOWNSTREAM

Data Lake · Data Warehouse · Dashboard · dbt

Tools in Use

Claude Code

20 skills · memory · multi-agent · Team environment AI

GitHub Copilot

Line completion · cross-model code review

NR MCP

Live logs · metrics · traces · Alert diagnosis

Jira + Confluence

Tickets + docs auto-updated every session

Code Review Graph

28 tools · blast radius · dependency impact analysis

Codeburn

Development analytics and metrics

RTK

Token optimisation · 60–90% Bash output savings

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DataOS Ingestion — How We Work Day to Day

Daily Workflow

1 Start with full context

Memory + codebase graph auto-load in seconds — zero manual setup

2 Copilot: lines + blind spot check

Autocomplete in IDE · cross-model review catches what Claude misses

3 Skills trigger automatically

/airflow · /pyspark · /nrql · /terraform · /sql · domain skills fire on context

4 Incidents diagnosed via NR MCP

30 min–3 hrs → 0–20 min for alert issues · 12 NR tools available in chat

5 Complex tasks via multi-agent

5 parallel agents — 2–3 days → 2–6 hours for big multi-file work

6 Wrap-up closes the loop

Jira auto-updated · knowledge captured · graph refreshed · 30 seconds

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Two AI Tools. Different Jobs. Better Together.

GitHub Copilot — In the IDE

- ✅ Autocomplete lines & functions as you type
- ✅ Inline chat: explain, refactor, or debug code
- ✅ Generate tests and boilerplate instantly
- ✅ Cross-model review: GPT catches Claude blind spots

Together: deliberate cross-model coverage —

Different AI models catch different blind spots

Claude leads · Copilot verifies · humans approve

+

Claude Code — Team Environment AI

- ✅ Full codebase context at every session start
- ✅ Queries NR, Jira, Confluence live in the chat
- ✅ Blast radius visible before writing a single line
- ⚡ **/avengers: 5 parallel agents — 2–3 days → 2–6h**
- ⚡ **Incident diagnosis via NR MCP — 0–20 min**
- ✅ 20 skills auto-trigger on file type or topic
- ✅ Jira + Confluence auto-update on every session end
- ✅ 5-layer review: CC Opus → Copilot GPT (blind spots) → agents → human

Together: 8–12× overall productivity for DataOS Ingestion DE tasks — reported by the team

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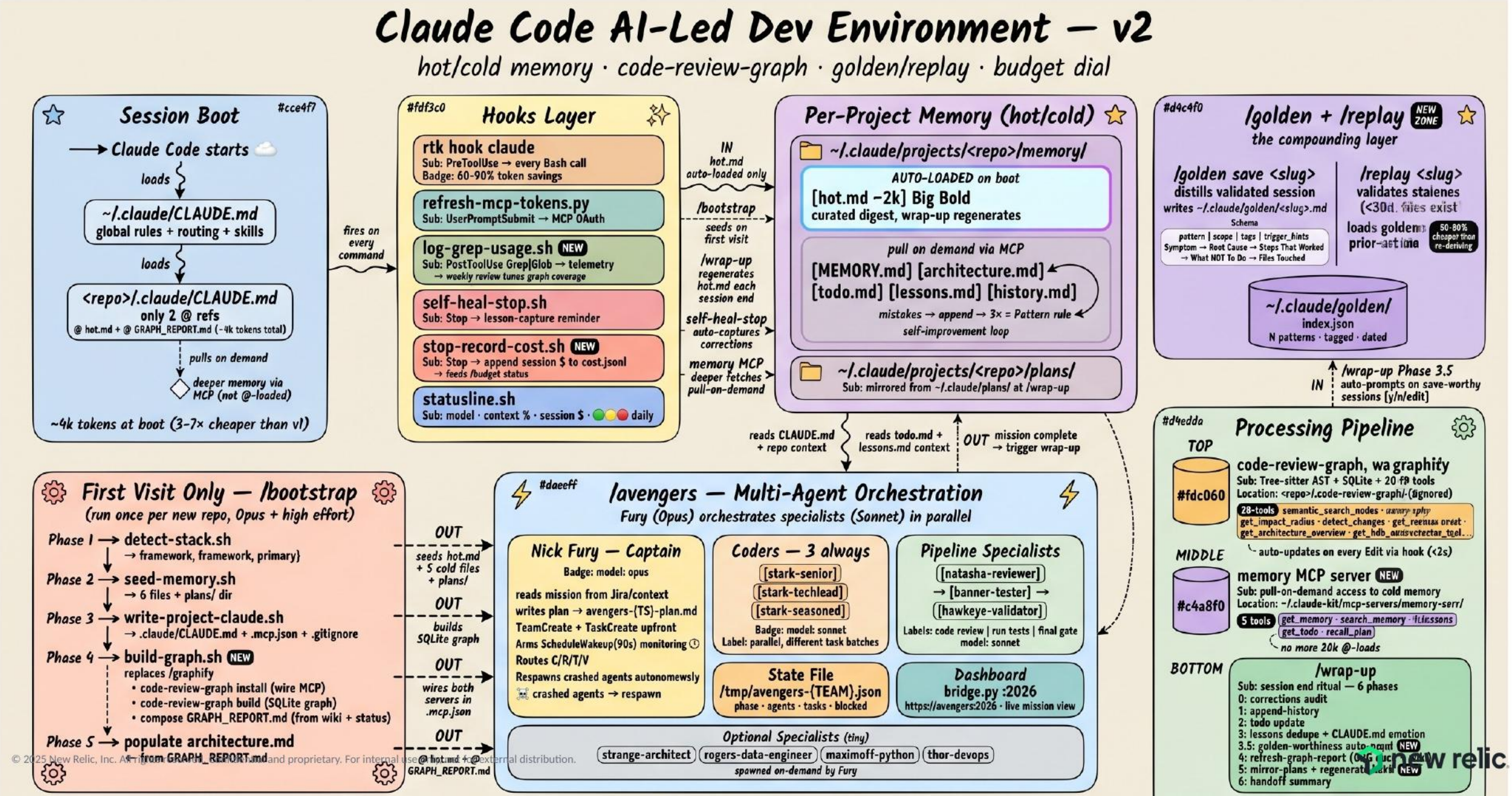
System Architecture — How It All Connects Under One Environment



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Full AI-Led Development Lifecycle — Time Savings at Every Phase

	01 Requirement	02 Spec & Plan	03 Architect	04 Build	05 Review	06 Test & Gate	07 Deploy	08 Monitor	09 Document	10 Learn
	Manager / PM	AI-assisted	AI + Graph	Task & volume dep.	5-layer AI+Human	AI auto-mated	Human initiates	NR MCP (AI)	AI auto-update	Golden patterns
Before:	—	4–6h	2–4d	2–3d	—	—	—	1–10h	manual	—
After:	—	20m–1h	1–2d	2–6h	—	—	—	0–20m	auto	auto

Confirmed by engineer · Spec+Arch+Build: days → hours · Alert issue diagnosis: hours → 0–20 min · Overall: 8–12×

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Faster Development — Days Became Hours

BEFORE

2–3 DAYS

Per development ticket — regardless of complexity
Context rebuilt from scratch every time

→

AFTER

2–6 HOURS

Task & volume dependent
Full context auto-loaded · skills auto-trigger · graph aware

Spec & Plan
4–6h → 20m–1h

AI assists planning from existing codebase patterns — not from scratch

Architecture
2–4d → 1–2d

Code graph shows impact + CDD & Confluence auto-updated

Context
always ready

Memory + graph auto-load
No archaeology every session

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/avengers — Five AI Engineers Working in Parallel on One Task

Tasks that take one engineer one full day, completed in under 2 hours:

Orchestrator (Opus model — plans, routes, validates, shuts down)

Planner	Coder ×N	Code Reviewer	Validator
Researches codebase → implementation plan	Parallel code writing verifies build passes	Code standards + security checks	Tests + impact analysis before any merge

1. Orchestrator reads spec → spawns Planner + Coders simultaneously
2. Each agent works on an independent file — zero waiting
3. Reviewer + Validator run in parallel on all output
4. Human sees a single clean PR to approve

1 DAY

↓

2 HOURS

Airflow DAG + dbt model
+ NR alert — all in parallel

Engineer reviews
1 PR instead of 5 files

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Incident Investigation — AI Queries New Relic Directly, Instantly

BEFORE — Manual hunt

2–3 HOURS

Log in to NR UI · navigate to entity · scroll logs
switch tabs · compare charts · Slack teammates

Form hypothesis → test → repeat → still uncertain
Customer impact growing every minute

→

AFTER — AI-powered diagnosis

30 MINS

"analyze kafka consumer lag last 4 hours"
Claude queries NR, returns structured analysis

Consumer lag · partition skew · throughput drop
Affected services identified — fix scoped immediately

12 New Relic tools · Jira auto-updated · root cause in minutes

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Impact Analysis — Know What Breaks Before Writing a Single Line

We see the full impact of every change BEFORE it touches production.

⚠ Before: Invisible blast radius

Engineer changes a pipeline file.
Deploys. A downstream consumer breaks silently.
5-day incident — all from one undetected dependency.

→

✅ After: Instant clarity

Impact analysis runs before coding.
Sees 3 callers, 1 downstream consumer.
Fix scoped. Ships confidently. Zero incident.

~1 sprint/qtr

absorbed in surprise rework — before

< 1 day

rework after impact-aware development

0 blind deploys

every change impact-checked first

Auto-updates

dependency map rebuilt on every file save

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Documentation That Writes Itself — Jira & Confluence Always Up to Date

Work session completes → /wrap-up fires (~30s) → Claude drafts summary → Jira ticket auto-updated → Confluence published

Jira — What gets auto-updated

Status	In Review → Done · automatically
Comment	What was done + decisions made
Worklog	Time logged from session duration
Links	Commits linked via branch name

Confluence — What gets published

Architecture	Decisions logged after every session
Postmortem	Incident page auto-created on wrap-up
Release	Notes from git history — auto-versioned
Runbooks	Updated as team learns new patterns

Zero manual effort. Every session documented. Every time.

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Knowledge That Compounds

Every session makes every future session smarter — knowledge never leaves the team

ALWAYS LOADED

Session memory — auto-loads at every session start

Active priorities + recent lessons + architecture summary loaded automatically. No manual setup. Zero tokens wasted.

PULL ON DEMAND

Team knowledge — retrieved only when relevant

Patterns · architecture decisions · session history · past plans. Fetched on demand — only what's needed, when it's needed.

REPLAYABLE WINS

Validated fixes — saved as reusable templates

After a confirmed fix: the steps are captured and indexed. Next time the same problem appears, the proven solution loads instantly.

Every session: team knowledge grows. Every fix: the next one is faster. The system gets smarter, not staler.

The Results — Numbers the Team Has Measured

8–12×

overall productivity
for DE tasks

2–6h

dev ticket
vs 2–3 days before

0–20m

Alert issue diagnosis
vs 1–10 hours before

25+ SP

sprint capacity
+ adhoc on top

More measurements

5–15 min

context reset after absence
(vs hours/days before)

3–5h

new pipeline (full cycle)
vs 2–4 days before

4–6h → 20m–1h

spec & plan phase

2–4d → 1–2d

architecture + CDD
+ Confluence

8–12× overall for DE tasks · All metrics reported directly by Ingestion Team engineers

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Before & After — Ingestion Team · 3 Repos · 3 Months

What We Measured	Before	After (AI-Led)
Spec & Plan	4–6 hours	**20 min – 1 hour** ✅
Architecture + CDD	2–4 days	**1–2 days** ✅
Development ticket	2–3 days	**2–6 hours (task-dependent)** ✅
New pipeline (full cycle)	2–4 days	**3–5 hours** ✅
Alert issue diagnosis	30 min – 3 hours	**0–20 minutes (NR MCP)** ✅
Context after absence	Hours / days	**5–15 min via /bootstrap** ✅
Sprint throughput	25 SP (at capacity)	**25 SP + additional + adhoc** ✅
Documentation effort	Manual, often skipped	**Automatic via /wrap-up** ✅
Code review layers	Manual review only	**CC Opus + Copilot + Natasha + Human** ✅

✅ = confirmed by Ingestion Team engineers · Repos: om-airflow-dags · spark-kafka-apps · tf-dataos-new-relic

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AI Does the Work. Humans Stay in Control. Always.

1	**Requirements**	Human writes the spec — AI reads it, asks clarifying questions
2	**Architecture**	Human approves the implementation plan — before a line is written
3	**Security Review**	AI flags every security vulnerability — any critical issue blocks the pipeline
4	**PR Merge**	Human reviews every PR — agent output treated like any other code
5	**Deploy**	Human initiates the deploy — AI writes release notes + monitors

Agent PRs treated identically to human PRs — same review, same standards, same gates.

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Getting Started — From Zero to Productive in 4 Steps

15 MIN

Step 1 — Install & Setup

One-time per engineer

Claude Code + MCP servers + Budget dial
Multi-orchestration + hooks + 20 skills
Fully automated — one script.

5 MIN

Step 2 — Bootstrap New Repo

First visit to any repo

Detects your stack, seeds team memory, builds codebase graph, wires all MCPs.
Runs once — context ready forever.

ALWAYS

Step 3 — AI-Led Development

Single agent or multi agent

Single task: Claude Code works alongside you.
Complex task: /avengers spawns parallel agents.
Copilot handles lines. Claude handles investigations.

30 SEC

Step 4 — Wrap-Up

Every session end

Jira auto-updated · lessons captured
Knowledge persisted · graph refreshed.
The habit that makes everything compound.

The Compounding Effect

Day 1

Full codebase context ready.
Development tickets: days → hours.

Month 1

Lessons accumulate. Patterns reused.
Alert diagnosis: hours → 0–20 min.

Month 3+

Knowledge never leaves the team.
8–12× productivity — sustained.

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What We're Asking

Support adoption

Drive AI-led workflows across all Data Engineering & Management in DTM.

Allocate time

Support the daily wrap-up habit — 30 seconds per session. Knowledge compounds automatically.

Endorse the model

Establish this as the standard template for all DE teams at New Relic.

8–12×

Productivity

2–6h

Dev ticket

0–20m

Alert diagnosis

25+ SP

Sprint capacity

Thank You · Questions?

Repo: github.com/nr-mdakilahmed/claude_code_setup